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Parasitism, mutualism and commensalism

Relationships between organisms take several forms. *Symbiosis* is the term used to describe how different plant or animal species live together. The effects of the relationship are then classified as *parasitism, mutualism* or *commensalism.*

Parasitism

The word *parasite* comes from an old word meaning 'one who eats at another's table'. A parasite is an organism that obtains its nutrients by living in or on another organism. The parasite obtains nutrients at the expense of the host organism it associates with. Thus, *parasitism* involves one organism depriving another of its nutrition. In this case, the parasite benefits and the host suffers. An example of a parasitic relationship is the tapeworm. Tapeworms can infect humans and other animals, living in the alimentary canal of the host. The tapeworm takes much of the nutrients its host ingests, leaving the host undernourished as a result (and in extreme cases, dead).

Mutualism

As the name suggests, *mutualism* involves a relationship that is of mutual benefit to both organisms. A well-known case of mutualism involves various species of birds living on or near larger animals, such as horses, cows, rhinos and giraffes. The birds eat the tiny bugs that would otherwise irritate or damage the skin of the larger animal. In exchange for providing this easy meal, the larger animal benefits by having its body kept free of pests. Similar examples occur in the oceans with smaller fish performing the same task on larger species. For example, a remora is a fish that has a disk on its head, which it uses like a sucker to cling onto other fishes, especially sharks. Remoras feed on food scraps left over by their hosts or on parasites of the host. They can also swim alone and have been seen clinging to the hulls of ships.

Commensalism

Commensalism describes a relationship in which one species benefits from another by way of obtaining food and/or shelter, but the other species receives no benefit at all. Commensalism differs from parasitism because the host organism in not harmed in any way.

1 Would you describe the relationship between a shark and a remora as *parasitism, mutualism* or *commensalism*. Explain your answer.

The digestive tracts of ruminants (ie cows, horses) contain cellulose-digesting bacteria.

2 Name the type of relationship the two species have.

3 How do the ruminants benefit from the relationship?

4 How do bacteria benefit from the relationship?

Lampreys and hagfish (*Class Agnatha*) are fishes with sucker-like mouths that attach to the host. Their rasping tongues are used to scrape away at the flesh of the fish, enabling them to feed on the tissue of their host.

5 How would you classify lampreys and hagfish? Explain your answer.

The common European cuckoo (*Cuculus canorus*) is a bird that lays its eggs in the nests of other birds which currently have eggs. The cuckoo's offspring are brought up by the original inhabitants of the nest – often causing their offspring to suffer.



6 How would you classify the cuckoo? Explain your answer.

Identify the terms defined by the following statements:

7 *In antagonistic symbiosis, this is the organism that suffers by satisfying the needs of another.*

8 *Hoofed mammals that chew the cud and have a complex four-chambered stomach.*

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